

Exercise – The Student's t distribution

1. Aim of the exercise

The aim of this exercise is to understand the t distribution using simulation.

2. Derivation of the distribution of the t statistic

In regression analysis, we are often interested in testing whether a population coefficient, such as β_1 , is equal to some hypothesized value β_1^0 . We formulate this as the null hypothesis and its alternative:

$$\begin{aligned} H_0 : \beta_1 &= \beta_1^0 \\ H_1 : \beta_1 &\neq \beta_1^0. \end{aligned}$$

We cannot test the null hypothesis directly because the population coefficient β_1 is unobserved. Instead, we use its sample estimate, $\hat{\beta}_1$, to draw inference about β_1 . Specifically, we assess whether the observed estimate $\hat{\beta}_1$ differs sufficiently from the hypothesized value β_1^0 to reject the claim that $\beta_1 = \beta_1^0$.

$\hat{\beta}_1$ is a random variable and has a sampling distribution. This means that there is a probability associated with observing different values of $\hat{\beta}_1$ relative to the hypothesized value β_1^0 . Consequently, the test of the null hypothesis must have a probabilistic nature. We therefore derive a test statistic that is itself a random variable and hence has a probability distribution.

The linear regression model in matrix form is given by

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}. \quad (1)$$

The vectors $\mathbf{y}$ and $\boldsymbol{\varepsilon}$ each contain n rows and one column. Let $\mathbf{X}$ be an $n \times K$ matrix, where each column contains the observations of one explanatory variable. Furthermore, let $\boldsymbol{\beta}$ be a $K \times 1$ column vector containing the coefficients $\beta_1, \beta_2, \dots, \beta_K$.

To derive the t -statistic in a general setting, we consider an arbitrary coefficient β_k , where $k = 1, \dots, K$. Its corresponding OLS estimate is denoted by $\hat{\beta}_k$, and the hypothesized value under the null hypothesis by β_k^0 .

The OLS estimator is given by

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}.$$

Substituting the linear regression model into the OLS estimator yields

$$\begin{aligned} \hat{\boldsymbol{\beta}} &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}) \\ &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}\boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon} \\ &= \boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\varepsilon}. \end{aligned}$$

If we assume

$$\boldsymbol{\varepsilon} \mid \mathbf{X} \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n),$$

then

$$\hat{\boldsymbol{\beta}} \mid \mathbf{X} \sim N \left[\boldsymbol{\beta}, \sigma^2 (\mathbf{X}'\mathbf{X})^{-1} \right].$$

That is, conditional on $\mathbf{X}$, the OLS estimator $\hat{\boldsymbol{\beta}}$ follows a Normal distribution with mean $\boldsymbol{\beta}$ and variance-covariance matrix $\sigma^2 (\mathbf{X}'\mathbf{X})^{-1}$. Since the test statistic is constructed from $\hat{\boldsymbol{\beta}}$, its

distribution is determined by the distribution of $\hat{\boldsymbol{\beta}}$. Because $\hat{\boldsymbol{\beta}}$ depends on the unobserved error term $\boldsymbol{\varepsilon}$, assumptions about the distribution of $\boldsymbol{\varepsilon}$ are needed. Here, we assume that $\boldsymbol{\varepsilon}$ follows a Normal distribution.

Subtracting $\boldsymbol{\beta}$ from both sides results in

$$\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} \mid \mathbf{X} \sim N \left[\mathbf{0}, \sigma^2 (\mathbf{X}'\mathbf{X})^{-1} \right].$$

The t -test is a hypothesis test conducted for a particular coefficient estimator contained in the vector $\hat{\boldsymbol{\beta}}$. Let $\hat{\beta}_k$ denote the k -th entry of $\hat{\boldsymbol{\beta}}$. From the distribution of $\hat{\boldsymbol{\beta}}$, it follows that

$$\hat{\beta}_k - \beta_k \mid \mathbf{X} \sim N \left[0, \sigma^2 S^{kk} \right],$$

where

$$S^{kk} \equiv \left[(\mathbf{X}'\mathbf{X})^{-1} \right]_{k,k}$$

denotes the k -th diagonal element of $(\mathbf{X}'\mathbf{X})^{-1}$. Under the null hypothesis, $\beta_k = \beta_k^0$. Substituting β_k^0 for β_k yields

$$\hat{\beta}_k - \beta_k^0 \mid \mathbf{X} \sim N \left[0, \sigma^2 S^{kk} \right].$$

The reason for this rewriting is that, conditional on $\mathbf{X}$,

$$\mathbb{E} \left[\hat{\beta}_k \mid \mathbf{X} \right] = \beta_k.$$

That is, $\hat{\beta}_k$ is centered around the true coefficient β_k . Under the null hypothesis, $\beta_k = \beta_k^0$, and therefore

$$\mathbb{E} \left[\hat{\beta}_k \mid \mathbf{X} \right] = \beta_k^0.$$

Consequently,

$$\mathbb{E} \left[\hat{\beta}_k - \beta_k^0 \mid \mathbf{X} \right] = 0.$$

The difference $\hat{\beta}_k - \beta_k^0$ therefore measures the extent to which the observed estimate deviates from what we would expect under the null hypothesis. This difference forms the basis of the t -test.

The next step is to standardize the distribution of $\hat{\beta}_k - \beta_k^0$. This is done by dividing by its standard deviation, $\sqrt{\sigma^2 S^{kk}}$. The purpose of this transformation is to obtain a distribution with mean zero and variance one, which allows us to compare the observed value to the standard Normal distribution. We therefore obtain

$$z_k \mid \mathbf{X} \equiv \frac{\hat{\beta}_k - \beta_k^0}{\sqrt{\sigma^2 S^{kk}}} \mid \mathbf{X} \sim N(0, 1).$$

The resulting distribution has mean zero and variance one. Because we have centered and scaled the distribution, it no longer depends on β_k^0 or σ^2 . Conditional on $\mathbf{X}$, z_k therefore follows the standard Normal distribution.

One remaining problem is that the true variance σ^2 is not observed and therefore cannot be used directly. Consequently, we replace σ^2 with its unbiased estimator

$$\hat{\sigma}^2 = \frac{\hat{\boldsymbol{\varepsilon}}'\hat{\boldsymbol{\varepsilon}}}{n - K}.$$

This yields the statistic

$$t_k = \frac{\hat{\beta}_k - \beta_k^0}{\sqrt{\hat{\sigma}^2 S^{kk}}}.$$

Due to the replacement of σ^2 by $\hat{\sigma}^2$, the distribution of t_k is no longer immediately known. To determine its distribution, we first rewrite the statistic in an equivalent form. This rewriting will allow us to derive the distribution of the numerator and denominator separately. We obtain

$$\begin{aligned} t_k &= \frac{\hat{\beta}_k - \beta_k^0}{\sqrt{\hat{\sigma}^2 S^{kk}}} \\ &= \frac{\hat{\beta}_k - \beta_k^0}{\sqrt{\hat{\sigma}^2} \sqrt{S^{kk}}} \frac{\sqrt{\sigma^2}}{\sqrt{\sigma^2}} \\ &= \frac{(\hat{\beta}_k - \beta_k^0) / \sqrt{\sigma^2 S^{kk}}}{\sqrt{\hat{\sigma}^2 / \sigma^2}} \\ &= \frac{(\hat{\beta}_k - \beta_k^0) / \sqrt{\sigma^2 S^{kk}}}{\sqrt{\hat{\sigma}^2 / \sigma^2}} \frac{1}{\sqrt{n-K} / \sqrt{n-K}} \\ &= \frac{(\hat{\beta}_k - \beta_k^0) / \sqrt{\sigma^2 S^{kk}}}{\sqrt{(\hat{\sigma}^2 / \sigma^2) (n-K) / (n-K)}}. \end{aligned}$$

We already know that the numerator of this fraction follows a standard Normal distribution:

$$\frac{\hat{\beta}_k - \beta_k^0}{\sqrt{\sigma^2 S^{kk}}} = z_k \sim N(0, 1).$$

Thus, conditional on $\mathbf{X}$, the numerator follows a standard Normal distribution.

We now derive the distribution of the denominator,

$$(\hat{\sigma}^2 / \sigma^2) (n - K).$$

Under the normality assumption, the standardized errors have distribution

$$\frac{\boldsymbol{\varepsilon}}{\sigma} \sim N(\mathbf{0}, \mathbf{I}_n).$$

A chi-squared random variable is defined as the sum of squares of independent standard Normal random variables. Consequently, the sum of squared standardized errors follows a chi-squared distribution. In regression analysis, the error variance is estimated by

$$\hat{\sigma}^2 = \frac{\hat{\boldsymbol{\varepsilon}}' \hat{\boldsymbol{\varepsilon}}}{n - K}.$$

Multiplying both sides by $(n - K) / \sigma^2$ yields

$$(\hat{\sigma}^2 / \sigma^2) (n - K) = \frac{\hat{\boldsymbol{\varepsilon}}' \hat{\boldsymbol{\varepsilon}}}{\sigma^2}.$$

Because K regression coefficients have already been estimated, only $n - K$ independent pieces of information remain in the residual vector $\hat{\boldsymbol{\varepsilon}}$. Therefore,

$$\frac{\hat{\boldsymbol{\varepsilon}}' \hat{\boldsymbol{\varepsilon}}}{\sigma^2} \sim \chi^2(n - K).$$

Hence,

$$(\hat{\sigma}^2/\sigma^2)(n-K) \sim \chi^2(n-K).$$

Substituting the distributions of the numerator and denominator into the rewritten expression for t_k yields

$$t_k \sim \frac{N(0,1)}{\sqrt{\frac{\chi^2[n-K]}{(n-K)}}} = t[n-K].$$

This expression provides the stochastic representation of the Student's t -distribution with $n-K$ degrees of freedom. It shows how a t -distributed random variable can be constructed from a standard Normal random variable and an independent chi-squared random variable.

Using this stochastic representation together with the density functions of the Normal and chi-squared distributions, it can be shown that the probability density function of the Student's t -distribution is

$$f(x) = \frac{\Gamma(\frac{\nu+1}{2})}{\sqrt{\pi\nu}\Gamma(\frac{\nu}{2})} \left(1 + \frac{x^2}{\nu}\right)^{-\frac{\nu+1}{2}},$$

where ν denotes the number of degrees of freedom. In our case, $\nu = n - K$.

Two gamma functions appear in the probability density function of the Student's t -distribution. Since the degrees of freedom satisfy $\nu = n - K > 0$, the arguments of the gamma function are strictly positive. For positive arguments, the gamma function can be expressed using Euler's integral:

$$\Gamma(z) = \int_0^\infty t^{z-1} e^{-t} dt, \quad z > 0.$$

4. Final notes

This file is prepared and copyrighted by Akash Boelens, Renata Istrătescu, and Tunga Kantarcı. This file and the accompanying MATLAB files are available on GitHub and can be accessed via this [link](#).